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漸化式の幸せな形

② B₁ $a_{n+1}-1 = 2(a_n-1)$, $a_1 = 2$ ← $a_{n+1} = 2a_n - 1$ とするのはもったいない

(解法1) $n \in n+1$ にした式 (解法2)

$$\begin{aligned} a_n - 1 = b_n \text{ とおくと } & \leftarrow a_{n+1} - 1 = b_{n+1} \quad \left(\{a_n - 1\} \text{ は } \textcircled{1} a_1 - 1, \textcircled{2} 2 \right. \\ & \left. \begin{cases} b_{n+1} = 2b_n \\ b_1 = a_1 - 1 = 1 \end{cases} \right. \left. \leftarrow \begin{cases} \{b_n\} \text{ は } \textcircled{1} 1, \textcircled{2} 2 \\ \text{の等比数列} \end{cases} \right) \end{aligned}$$

であるから

$$b_n = 1 \cdot 2^{n-1} = 2^{n-1}$$

$$a_n - 1 = (a_1 - 1) \cdot 2^{n-1}$$

$$a_n - 1 = b_n \text{ より}$$

$$= 2^{n-1}$$

$$a_n = b_n + 1$$

$$\therefore a_n = 2^{n-1} + 1 //$$

$$= 2^{n-1} + 1 //$$

② B₂ $a_{n+1} + 2(n+1) + 1 = 3(a_n + 2n + 1)$, $a_1 = 1$

(解法1) $n \in n+1$ にした式 (解法2)

$$\begin{aligned} a_n + 2n + 1 = b_n \text{ とおくと } & \leftarrow a_{n+1} + 2(n+1) + 1 = b_{n+1} \quad \left(\{a_n + 2n + 1\} \text{ は } \textcircled{1} a_1 + 2 \cdot 1 + 1 \right. \\ & \left. \begin{cases} b_{n+1} = 3b_n \\ b_1 = a_1 + 2 \cdot 1 + 1 = 4 \end{cases} \right. \left. \leftarrow \begin{cases} \{b_n\} \text{ は } \textcircled{1} 4, \textcircled{2} 3 \\ \text{の等比数列} \end{cases} \right) \end{aligned}$$

であるから

$$b_n = 4 \cdot 3^{n-1}$$

$$a_n + 2n + 1 = (a_1 + 2 \cdot 1 + 1) \cdot 3^{n-1}$$

$$a_n + 2n + 1 = b_n \text{ より}$$

$$= 4 \cdot 3^{n-1}$$

$$a_n = b_n - 2n - 1$$

$$\therefore a_n = 4 \cdot 3^{n-1} - 2n - 1 //$$

$$= 4 \cdot 3^{n-1} - 2n - 1 //$$

② B₃ $a_{n+1} - 2 \cdot 3^{n+1} = 5(a_n - 2 \cdot 3^n)$, $a_1 = 7$

(解法1) $n \in n+1$ にした式 (解法2)

$$\begin{aligned} a_n - 2 \cdot 3^n = b_n \text{ とおくと } & \leftarrow a_{n+1} - 2 \cdot 3^{n+1} = b_{n+1} \quad \left(\{a_n - 2 \cdot 3^n\} \text{ は } \textcircled{1} a_1 - 2 \cdot 3^1 \right. \\ & \left. \begin{cases} b_{n+1} = 5b_n \\ b_1 = a_1 - 2 \cdot 3^1 = 1 \end{cases} \right. \left. \leftarrow \begin{cases} \{b_n\} \text{ は } \textcircled{1} 1, \textcircled{2} 5 \\ \text{の等比数列} \end{cases} \right) \end{aligned}$$

であるから

$$b_n = 1 \cdot 5^{n-1} = 5^{n-1}$$

$$a_n - 2 \cdot 3^n = (a_1 - 2 \cdot 3^1) \cdot 5^{n-1}$$

$$a_n - 2 \cdot 3^n = b_n \text{ より}$$

$$= 5^{n-1}$$

$$a_n = b_n + 2 \cdot 3^n$$

$$\therefore a_n = 5^{n-1} + 2 \cdot 3^n //$$

$$= 5^{n-1} + 2 \cdot 3^n //$$

point

漸化式の幸せな形とは

$$\boxed{\phantom{a_{n+1}}} = \bigcirc \times \boxed{}$$

$n \in n+1$ にした式